

HPOS — Factory Operating System

~ proposal by saneee.in [29/08/2026]

Digital Transformation & Business Management Platform for Himalaya Plast

Self-Hosted VPS

Transitioning Himalaya Plast from disconnected WhatsApp chats, manual Excel sheets, and paper QC logs into a single, unified factory operating system with 100% end-to-end extrusion traceability.

01. SYSTEM ARCHITECTURE & TECH STACK

BACKEND CORE (70%)

ERPNext v15 + Frappe Engine

The central industrial backbone acting as the **single source of truth** for financial, MRP, and compliance data.

- **Manufacturing & MRP:** Multi-level BOMs, Work Orders, & Job Cards.
- **Batch Genealogy:** Full backward/forward RM-to-FG tracking.
- **Accounting & Tax:** Invoicing, Payments, P&L, and General Ledger.
- **Statutory GST:** Native e-Invoice (IRN/QR) & E-Way Bill generation.

CUSTOM UI LAYER (30%)

Custom Next.js & React Web Portal

A tailored, high-speed custom frontend communicating with ERPNext via secured REST APIs for shop floor use.

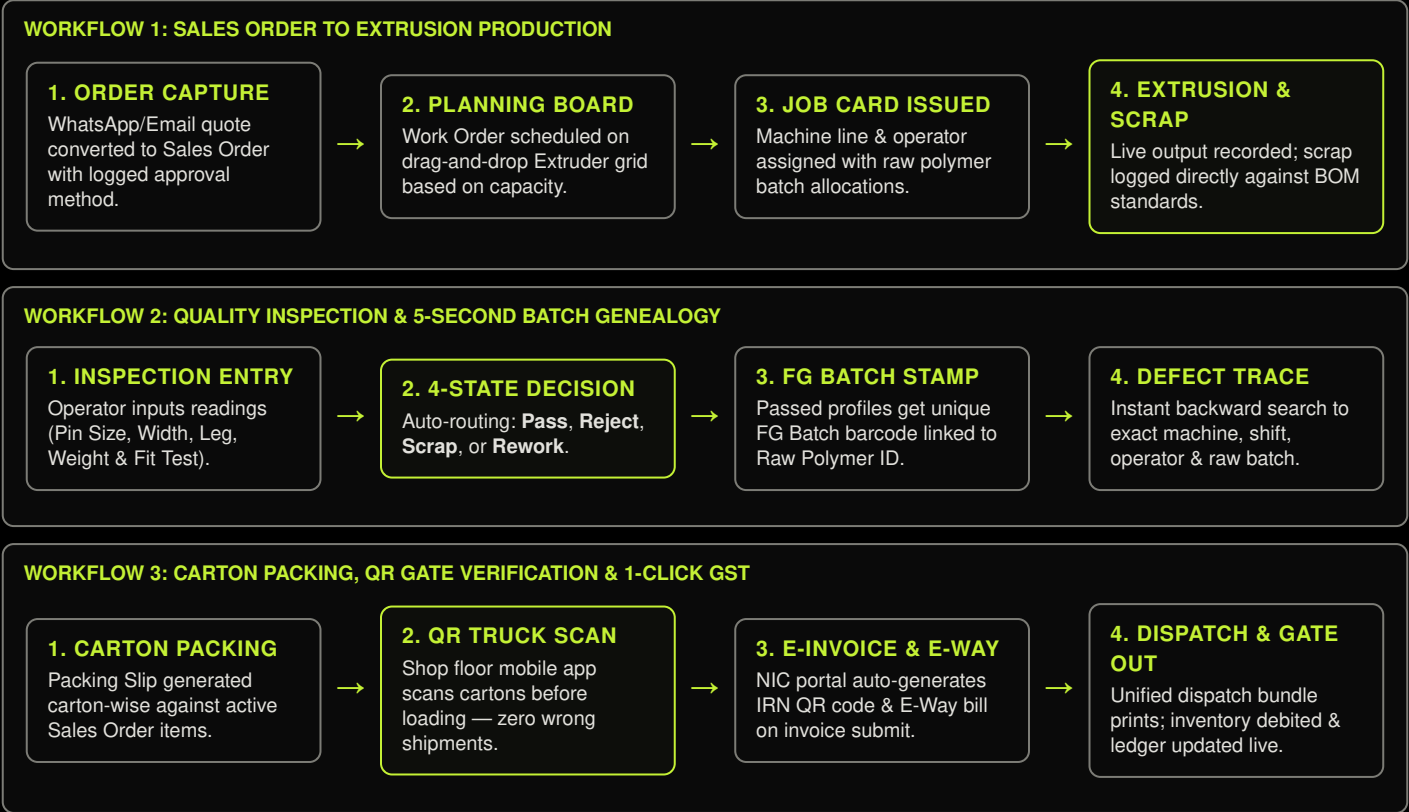
- **Founder Dashboard:** Live P&L, operational runway & risk orders.
- **Visual Planning Board:** Drag-and-drop machine scheduling grid.
- **Shop Floor QR Scanner:** Instant carton & dispatch verification.
- **10-Stage Timeline:** End-to-end Order-to-Cash tracking.

02. CORE BUSINESS WORKFLOWS WE ARE DIGITIZING

OPERATIONAL MODULE	CURRENT PROBLEM / MANUAL GAP	HPOS DIGITAL SOLUTION
Sales & Orders	Quotations and approvals lost across WhatsApp and phone calls.	Standardized quotes with multi-channel approval audit logs.
Extrusion Planning	Machine scheduling stored in heads, causing bottleneck clashes.	Visual Planning Board assigning Work Orders to specific extruders.
Quality Control	Paper QC sheets; zero digital history or defect trend analytics.	Digital QC templates (Pin, Width, Leg, Weight, Fit) with 4-state workflow.
Batch Traceability	Days spent manually investigating defective customer batches.	5-second batch genealogy from Delivery Note back to Raw Polymer batch.
Packing & Dispatch	Manual document drafting and missed cartons during truck loading.	Carton-wise packing slips with mobile QR scanning verification.
GST & Compliance	Double entry in external software for e-Invoices and E-Way bills.	1-click auto-generation of IRN, QR code, and E-Way bills directly.

03. TARGET SYSTEM WORKFLOWS — HOW YOUR FACTORY OPERATES WITH HPOS

Here is how HPOS transforms your daily operations across production, quality verification, and gate dispatch:



04. EXECUTIVE & SHOP-FLOOR VISIBILITY SUMMARY



05. PROJECT IMPLEMENTATION ROADMAP

Structured agile rollout ensuring your factory operates on a live baseline system within **8 weeks**, followed by custom extensions.

0

Phase 0: Setup (W1–3)

VPS Linux environment setup, Dockerized ERPNext configuration, Chart of Accounts, Item Masters & user permission matrix.

1

Phase 1: Core MRP (W3–16)

Deploy Order-to-Cash, BOMs, Job Cards, Inventory, QC workflows & India GST Compliance. *Working live core by ~Week 8.*

2

Phase 2: Custom UI (W16–24)

Custom Next.js interface rollout: Founder Dashboard, 10-Stage Order Tracking Stepper, and Shop-Floor QR Scanning tool.

3

Phase 3: Planning (W24–32)

Visual Drag-and-Drop Machine Planning Grid built using live production cycle-time data and capacity benchmarks.

06. WHAT WE NEED FROM YOU

- **Company Masters:** GSTIN, plant addresses, bank details, & Chart of Accounts.
- **Item Catalog & BOMs:** List of polymers, masterbatches, and profile formulas.
- **Machine & Station List:** Extrusion lines, output rates, and assigned operators.
- **QC Inspection Parameters:** Tolerance bands for Pin, Width, Leg, Weight & Fit.
- **Opening Balances:** Raw material stock, finished goods, and vendor/customer ledgers.

07. EXPECTED OUTCOMES & ROI

- **100% Traceability:** Reverse trace customer defects to the exact machine & batch in 5s.
- **Zero Double Entry:** Invoicing, e-Invoice & E-Way bills generated from one screen.
- **Live Visibility:** Real-time overview of open orders, machine load, and receivables.
- **Error-Free Dispatches:** Barcode/QR verification guarantees zero wrong cartons loaded.

08. PROJECT DELIVERABLES SUMMARY

Production ERP Core

ERPNext v15 + GST Compliance

Custom Next.js Web App

Dashboard, Scanner & Timeline

Dedicated VPS Setup

Dockerized Cloud Deployment